

Algorithmic Construction of Weak Orders between Translation-Invariant Total Orders (TITO)

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Last update July 17, 2026

Background: Partial Order

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Definition [1]

A partially ordered set is set P together with a relation $\preceq$ on P that satisfies the following axioms

- 1 $a \preceq a, \forall a \in P$ (reflexivity).
- 2 For all $a, b \in P$, if $a \preceq b$ and $b \preceq a$, then $a = b$ (antisymmetry).
- 3 For all $a, b, c \in P$, if $a \preceq b$ and $b \preceq c$, then $a \preceq c$ (transitivity).

Partial order $\preceq$ is a **total order** if for all $a, b \in P$, $a \preceq b$ or $b \preceq a$ (comparability).

Background: TITO

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Definition [2]

Fixed a positive integer n . A translation-invariant total order (TITO) is a total order $\preceq$ on $\mathbb{Z}$ such that for all $a, b \in \mathbb{Z}$, $a \preceq b$ if and only if $a + n \preceq b + n$.

- Size of block: number of residue classes modulo n appearing in that block.
- Block I is a **waxing** block if $x \preceq x + n$ for all $x \in I$ and I is a **waning** block if $x + n \preceq x$ for all $x \in I$.

Background: Window

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- Window: encode data of a block.
- Window notation for TITO $\preceq$ consists of a window for each block, ordered from left to right as in $\preceq$.

Example: $n = 3$

Example of TITO:

$\dots 4 \preceq 1 \preceq -2 \preceq -5 \dots -3 \preceq 2 \preceq 0 \preceq 5 \preceq 3 \preceq 8 \dots$

Blocks:

1 $[\dots 4 \preceq 1 \preceq -2 \preceq -5 \dots] \implies \text{window: } [\underline{1}]$

2 $[\dots 2 \preceq 0 \preceq 5 \preceq 3 \dots] \implies \text{window: } [2, 0]$

Window notation: $[\underline{1}][2, 0]$

Background: Weak Order

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Reflection Index

Consider equivalence relation $\sim$: $(a, b) \sim (a + n, b + n)$.

Denote the equivalence class of (a, b) as (a, b) . A reflection index is an equivalence class (a, b) such that $a < b$.

- imaginary reflection index ($n = 3$): $(3, 6)$
- real reflection index ($n = 3$): $(3, 4)$

Definition: Weak Order

Inversion: An *inversion* of a TITO $(\prec)$ is a reflection index (a, b) such that $b \prec a$. Let $N(\prec)$ denote the set of inversions of $(\prec)$.

Weak order: We define a partial order on TITOs by setting

$$(\prec_1) \leq (\prec_2) \quad \text{if and only if} \quad N(\prec_1) \subseteq N(\prec_2).$$

Example of Inversion Set

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Example: $n = 3$

Consider the following TITOs:

$$1 \quad \cdots \prec -2 \prec -1 \prec 0 \prec 1 \prec 2 \prec 3 \prec \cdots = [1, 2, 3]$$

$$N(\prec_1) = \emptyset$$

$$2 \quad \cdots \prec 3 \prec 2 \prec 1 \prec 0 \prec -1 \prec -2 \prec \cdots = [\underline{3}, 2, 1]$$

$$N(\prec_2) = \{(a, b) : a < b\}$$

$$3 \quad \cdots 4 \preceq 1 \preceq -2 \cdots -3 \preceq 2 \preceq 0 \preceq 5 \preceq 3 \preceq 8 \cdots = [\underline{1}][2, 0]$$

$$N(\prec_3) = \{(1, 4), (1, 7), (1, 10) \cdots\} \cup \{(0, 2)\} \cup \\ \{(2, 4), (2, 7), \cdots\} \cup \{(0, 1), (0, 4), (0, 7) \cdots\}$$

Motivation

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- The resulting poset of the TITOs is denoted $\text{WO}(\text{TTot}_n)$, a combinatorial model for the extended weak order of the affine symmetric group, denoted $\text{Dyer}(\tilde{S}_n)$.
- Understanding how two TITOs compare in this poset can help revealing the structure of the extended weak order and its lattice properties.
- Developing an algorithm to compare TITOs provides a concrete and computational way to study and visualize this structure.

Goal of the Project

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- The weak order between two TITOs are hard to compare, since the inversion sets can be infinite.
- The goal of our project is to design algorithms to determine the weak order between two TITOs.
- One possible roadmap is to come up with finite representation of the inversion set by further investigating its structure.

Motivations behind Normalizing Window Notations

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- A translation-invariant total order (TITO) on $\mathbb{Z}$ can be represented by *window notation*:

$$[x_1, x_2, \dots, x_n], \quad x_i \in \mathbb{Z}, \quad x_i \not\equiv x_j \pmod{n}.$$

- Different windows may encode the *same* TITO, for instance when $n = 3$:

$$[1, 2, 3] \equiv [4, 5, 6] \quad (\text{translation by } +3).$$

- **Solution:** Apply a *normalization algorithm* so every TITO has a unique normalized window such that
 - the number with the smallest residue class is the start of the normalized block.
 - the first number is an integer between 0 and $n - 1$

Step 1: Compute Residues

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Goal: Compute each element's residue and identify the smallest residue class.

- Compute $r_i = x_i \bmod n$ for each x_i in $[x_1, \dots, x_n]$.
- Find the index i such that r_i is the smallest residue.

Example

For $W = [4, 5, 6]$ and $n = 3$:

$$r_i = [1, 2, 0].$$

The smallest residue is 0, corresponding to $x_i = 6$.

Step 2: Shift for Smallest Residue at Window's Start

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Goal: Move the element with the smallest residue to the front of the window.

- Shift the block so that x_i (with smallest residue) comes first:

$$[x_i, x_{i+1}, \dots, x_n, x_1 + n, \dots, x_{i-1} + n].$$

Example

Smallest residue corresponds to $x_i = 6$:

$$[6, 4 + 3, 5 + 3] = [6, 7, 8].$$

Step 3: Compute Translation Constant

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Goal: Find the integer k such that all residues align properly.

- Compute

$$k = x_i - r_i,$$

where x_i is the element with smallest residue r_i .

Example

From $x_i = 6$, $r_i = 0$:

$$k = 6 - 0 = 6.$$

Step 4: Normalize by Subtracting k

Goal: Subtract k from each entry to get the normalized window.

- For each x_j in the shifted block, compute $y_j = x_j - k$.
- The result $[y_1, \dots, y_n]$ is the normalized window, satisfying:

$$\min_i y_i = \min_i r_i, \quad y_i \equiv r_i \pmod{n}.$$

Example

Subtract $k = 6$ from $[6, 7, 8]$:

$$[6, 7, 8] - [6, 6, 6] = [0, 1, 2].$$

Structure of the Inversion Set

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- Here we take TITO with $n=3$ as an example.
- **Inversion Set** consists of two types of reflection indices:
 - **Imaginary reflection index**
 - For residue class 0
 - For residue class 1
 - For residue class 2
 - **Real reflection index**
 - Residue classes 0 and 1 $\rightarrow$ 6 cases
 - Residue classes 0 and 2 $\rightarrow$ 6 cases
 - Residue classes 1 and 2 $\rightarrow$ 6 cases

Comparison Algorithm Overview

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Given fixed n , we want to compare two TITOs' orders $(\prec_1)$ and $(\prec_2)$ with possible outputs:

$$(\prec_1) \not\preceq (\prec_2), \quad (\prec_2) \not\preceq (\prec_1), \quad (\prec_1) = (\prec_2), \quad \text{incomparable.}$$

Our Example

When $n = 3$,

$$\prec_1 = [0, 2, 7]$$

$$\prec_2 = [2][\underline{0}, -5]$$

Imaginary Reflection Indexes

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Special Case: same residue class

Only “all or none” possibilities occur! (i.e., waxing or waning)

Our Example

<i>residue classes</i>	0	1	2
$\prec_1 = [0, 2, 7]$	<i>none</i>	<i>none</i>	<i>none</i>
$\prec_2 = [2][0, -5]$	<i>all</i>	<i>all</i>	<i>none</i>

$n = 3$	0	1	2
0	<		
1		<	
2			=

Real Reflection Indexes: Special Block Cases

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Consider residue class i with j , where $i < j$. When $n = 3$, and let us compare the residue class of 0 and 1!

Special cases

- $[0, 1] \Rightarrow \emptyset$
- $[0, -2] \Rightarrow$ all: $\{(0, j') \text{ for all } j' \text{ in residue class } 1\}$

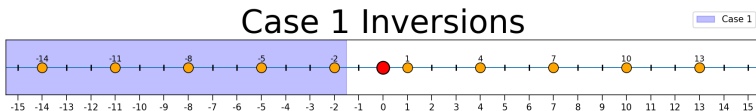
Real Reflection Indexes: Case 1

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Case 1: Different block $[0][1]$

$$\{(-2, 0), (-5, 0), (-8, 0), \dots\}$$



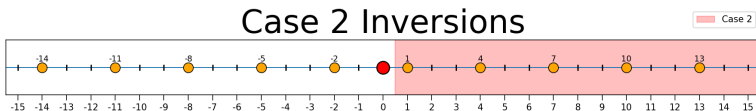
Real Reflection Indexes: Case 2

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Case 2: Different block $[1][0]$

$$\{(0, 1), (0, 4), (0, 7), \dots\}$$



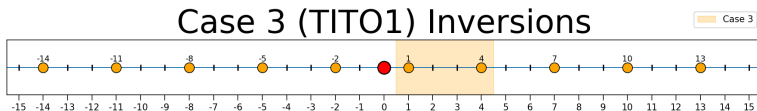
Real Reflection Indexes: Case 3

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Case 3: $[0, 7], 0 < 7 \rightarrow$ finite inversion set.

$$\{(0, 1), (0, 4)\}$$



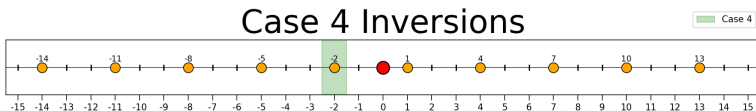
Real Reflection Indexes: Case 4

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Case 4: $[0, -2], 0 > -2 \rightarrow$ finite inversion set.

$$\{(0, -2)\}$$



Real Reflection Indexes: Case 5

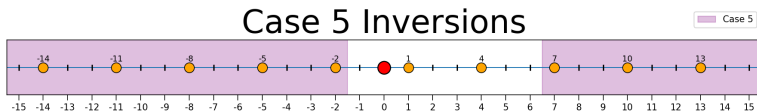
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Case 5: $[0, 4]$, $0 < 4 \rightarrow$ cofinite inversion set.

Reverse of $[4, 0]$

$$\{(0, 1), (0, 4)\}^C$$



Real Reflection Indexes: Case 6

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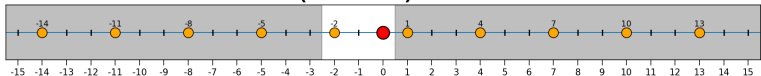
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Case 6: $[0, -5], 0 > -5 \rightarrow$ cofinite inversion set.

Reverse of $[-5, 0]$

$$\{(0, -2)\}^C$$

Case 6 (TITO2) Inversions



Real Reflection Indexes: Comparing Cases

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Inversion set inclusions:

$$3 \subset 2 \subset 6, \quad 4 \subset 1 \subset 5.$$

- Cases (1,4,5) may be comparable.
- Cases (2,3,6) may be comparable.
- Between the two groups $\rightarrow$ incomparable.

Our Example

Comparing residue class pair 0&1,

	case
$\prec_1 = [0, 2, 7] \Rightarrow [0, 7]$	3
$\prec_2 = [2][\underline{0}, -5] \Rightarrow [\underline{0}, -5]$	6

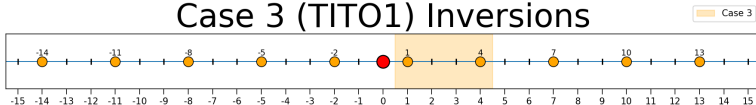
where for this pair, $\{\text{inversions of } \prec_1\} \subset \{\text{inversions of } \prec_2\}$.

Real reflection indexes: Visualization

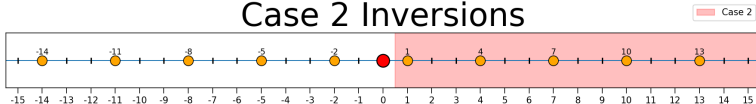
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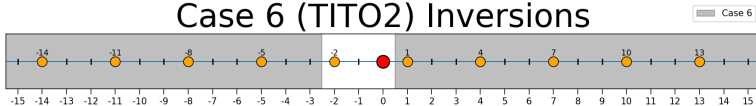
Case 3 (TITO1) Inversions



Case 2 Inversions



Case 6 (TITO2) Inversions

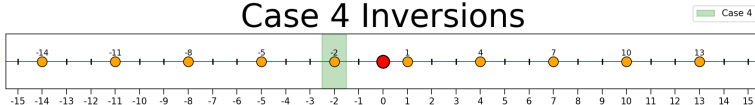


Real reflection indexes: Visualization (Cont.)

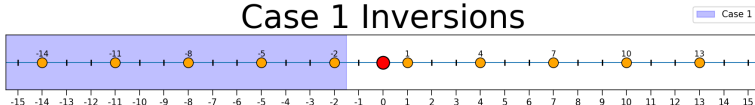
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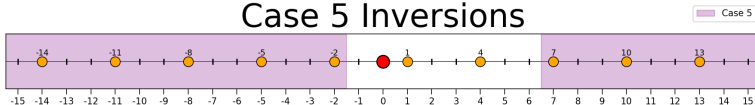
Case 4 Inversions



Case 1 Inversions



Case 5 Inversions



Final Decision Rules

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We construct a table for each residue class pair with outputs:

$<, >, =, x$

(subset, superset, equal, incomparable).

$n = 3$	0	1	2
0	$<$	$<$	$<$
1		$<$	x
2			$=$

- 1 If any x , overall incomparable, so our example is incomparable!
- 2 If both $<$ and $>$, overall incomparable.
- 3 If $<$ and rest $=$, then $(\prec_1) \not\preceq (\prec_2)$.
- 4 If $>$ and rest $=$, then $(\prec_2) \not\preceq (\prec_1)$.
- 5 Otherwise all equal (ruled out by normalization).

Implementation

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■ Data structures

- Block: a block of integers (constructor from_list used in tests).
- TIT0: holds modulus n and a list of Blocks.

■ Core functions

- `classify_pair(...)`: classify a concrete ordered pair into a case.
- `compare_pairs(t1, t2)`: aggregate pairwise results per residue pair.
- `tito_relation_full(n, t1, t2) → (table, verdict)`.

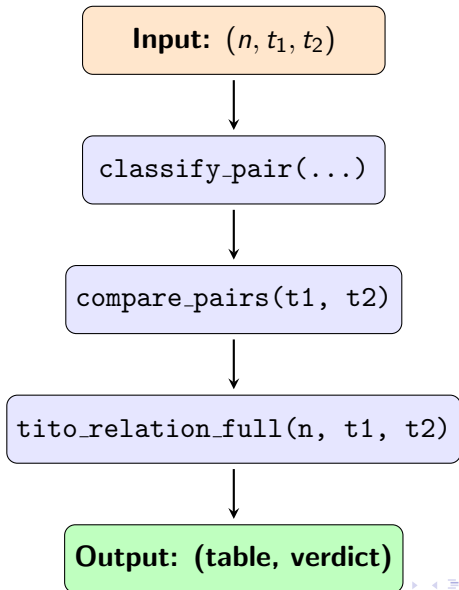
■ Outputs

- table: a mapping on residue pairs (i, j) with cell results.
- verdict: one of {"equal", " $t1 \not\preceq t2$ ", " $t2 \not\preceq t1$ ", "incomparable"}.

Workflow

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Testing and Performance

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■ Current status

- Unit tests in `test.py` pass.

■ Planned additions

- *Modulus mismatch*: specify behavior when `t1.n != t2.n` (error or 'incomparable') and add a test.
- *Boundary blocks*: singleton blocks and mixed small cases to ensure no missing cells or unexpected keys.

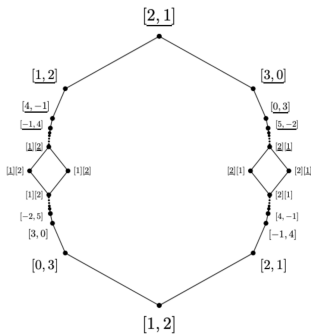
■ Current cost

- We generate many ordered pairs: grows roughly with the square of total elements.
- The result table only stores the cells we actually touch, the cost will be $O(n^2)$ (at most n^2).

Next Steps

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$n=2$



?

$n>2$

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Special thanks to our mentors, Grant Barkley and Mia Smith!